

Interface Control Document

Mk1 interfaces IF-01 to IF-22: definitions, timing, error handling and verification

Document	BOATY-ICD-001
Issue	Issue B (for review)
Date	28 September 2026
Status	For review by the project owner
Parent	BOATY-ADD-001 Issue C (interface register, section 5)
Content	22 interfaces, 29 verification entries, 13 open items to be confirmed

Document control

Issue	Date	Change	By
A	28 September 2026	First issue, for review.	Claude (drafted)
B	28 September 2026	TBC-10 closed: TALK button approved and defined in IF-12.	Claude, owner decision

Role	Name	Signature / date
Author	Claude, for Project Boaty	
Reviewer / approver	Project owner	

Review guidance: check that each interface is complete enough to build and test against. Items marked [TBC-nn] are known gaps with an owner; they're listed in section 9.

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1. Introduction

1.1 Purpose and scope

This document defines every Mk1 interface in the ADD's interface register in enough detail to build, integrate and test each side independently. For each interface it gives the owner, the definitions (signals, messages, fields, dimensions), the timing and error behaviour, and how the interface is verified. Interfaces internal to a third-party product (ArduPilot internals, the Claude API itself) are referenced, not re-specified.

1.2 Ownership

Each interface has one owning subsystem. The owner maintains the definition here and approves changes. The other side conforms. A change to a definition goes through an ICD revision before implementation, and the interface's verification is re-run.

1.3 References

Ref	Document
[1]	BOATY-SRS-001 Issue D; BOATY-ADD-001 Issue C
[2]	MAVLink common message set and mission protocol (mavlink.io)
[3]	ArduPilot Rover documentation: modes, failsafes, fence, SITL (ardupilot.org/rover)
[4]	Claude API documentation: Messages API, structured outputs, refusals and fallbacks (platform.claude.com)
[5]	RFC 7946 GeoJSON; RFC 8259 JSON; IEEE 802.11n
[6]	mavlink-router (github.com/mavlink-router)

2. Common conventions

Topic	Convention
Units	SI. Speed m/s, distance m, angles degrees (0-360, true north, clockwise), voltage V, current A, energy Wh
Position	WGS84 decimal degrees. MAVLink integers are degrees × 1e7. GeoJSON order is [lon, lat].
Time	UTC everywhere; ISO 8601 strings in JSON; GNSS time where available (SWE-008)
Text	UTF-8 JSON, no trailing commas, unknown fields ignored within a major version
Versioning	Schemas carry 'schema': 'boaty.<name>/<major>'. APIs are versioned in the path (/v1).
Logging	Every interface failure is logged with a UTC timestamp and interface ID (LOG-002)

3. Interface summary

ID	Interface	Between	Type	Owner	Section
IF-01	Boat-bank radio link	MCN ↔ MCP	Data	MCN	4
IF-02	Command & telemetry	MCN ↔ HLM	Data	MCN	4
IF-03	Photo & health service	MCN ↔ MCP	Data	MCP	4
IF-04	Companion link	MCP ↔ HLM	Data	MCP	4
IF-05	Motor commands	HLM ↔ PRP	Data	HLM	6
IF-06	Motor power	PWR ↔ PRP	Power	PWR	6
IF-07	Helm power & sensing	PWR ↔ HLM	Power	PWR	6
IF-08	Mission computer power	PWR ↔ MCP	Power	PWR	6
IF-09	Beacon drive	HLM ↔ REC	Data	HLM	6
IF-10	Internet uplink	MCN ↔ Phone	Data	MCN	4
IF-11	Claude API	MCN ↔ Claude API	Data	MCN	4
IF-12	Crew & operator panel	MCN ↔ Operator & crew	HMI	MCN	7

ID	Interface	Between	Type	Owner	Section
IF-13	Mission format	MCN	Software	MCN	5
IF-14	Helm API	MCN	Software	MCN	5
IF-15	Site data	MCN ↔ SIM	Software	MCN	5
IF-16	Hull segment joint	HUL	Mechanical	HUL	8
IF-17	Pod mount	PRP ↔ HUL	Mechanical	PRP	8
IF-18	Box & glands	PWR ↔ HUL	Mechanical	HUL	8
IF-19	Mast socket	HLM ↔ HUL	Mechanical	HUL	8
IF-20	DUPLO deck	HUL ↔ Operator & crew	Mechanical	HUL	8
IF-21	Simulation link	SIM ↔ MCN	Data	SIM	5
IF-22	GNSS & compass	HLM	Data	HLM	6

4. Network and data interfaces

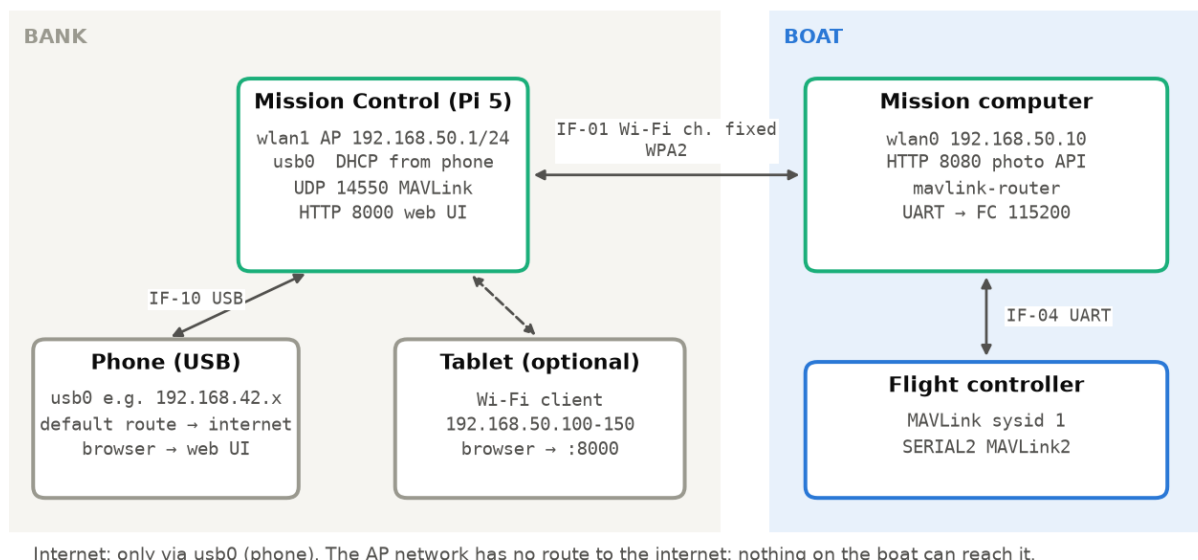


Figure 1. Network topology and addressing (IF-01, IF-04, IF-10).

IF-01 Boat-bank radio link

Interface	IF-01 Boat-bank radio link	Owner	MCN (Mission Control)
Between	Mission Control ↔ Mission computer	Type	Data
Medium	Wi-Fi 2.4 GHz 802.11n, WPA2, Pi 5 is AP	Requirements	COM-001..006

Radio and link layer

Parameter	Value
Standard / band	IEEE 802.11n, 2.4 GHz, 20 MHz channel
Channel	Fixed; chosen on site from 1, 6 or 11 by a scan at set-up (least occupied). Stored per site.
Access point	Mission Control USB adapter (wlan1), hostapd. Pi 5 internal Wi-Fi unused.
SSID / security	boaty-mc (hidden: no) / WPA2-PSK (WPA3-SAE if both ends support it), 20+ character passphrase, not the default
Transmit power	Default regulatory maximum for GB (country code GB in hostapd and wpa_supplicant)

Parameter	Value
Antennas	Bank: ≈ 5 dBi omni on the adapter, optional 2 m pole. Boat: Pi Zero 2W on-board antenna, box lid ≥ 150 mm above water.

Addressing (IPv4)

Host	Interface	Address	Notes
Mission Control	wlan1	192.168.50.1/24	DHCP server for .100-.150; no default route on wlan1
Mission computer	wlan0	192.168.50.10	Static; hostname boaty-mcp
Tablet (optional)	Wi-Fi	192.168.50.100-150	DHCP
Mission Control	usb0	Phone-assigned	See IF-10

Ports and traffic

Flow	Protocol / port	Direction	Rate
MAVLink (IF-02)	UDP 14550 on Mission Control	Router on boat is the client	$\approx 2\text{-}4$ kB/s
Photo & health API (IF-03)	HTTP TCP 8080 on MCP	Mission Control is the client	Bursty, up to link rate
Web UI (IF-12)	HTTP TCP 8000 on Mission Control	Tablet/phone is the client	Low

Security and error handling

- The MCP firewall (nftables) accepts inbound TCP 8080 only from 192.168.50.1 and drops everything else. MAVLink leaves the boat only towards 192.168.50.1:14550.
- Link loss is expected and never unsafe. Behaviour is defined by FS-002/003 on the helm and B4 on the MCP. Mission Control shows 'link lost' within 3 s (no HEARTBEAT) and keeps the last known position (MC-009).
- Signal quality: MCP reports its RSSI in /v1/health (IF-03). Mission Control also logs the AP's per-station RSSI once a second (COM-004).

Verification

Method	Stage	Evidence / pass criterion
Test	BENCH	Association, addressing and firewall rules checked with a script; ports reachable only as listed
Test	LAKE	≥ 100 m over water with telemetry ≥ 1 Hz and command latency ≤ 1 s at the 95th percentile (COM-002/003, V-08)

IF-02 Command & telemetry

Interface	IF-02 Command & telemetry	Owner	MCN (Mission Control)
Between	Mission Control $\leftrightarrow$ Helm	Type	Data
Medium	MAVLink 2 over UDP, relayed by B1	Requirements	COM-003, VAL-010, MOD-006

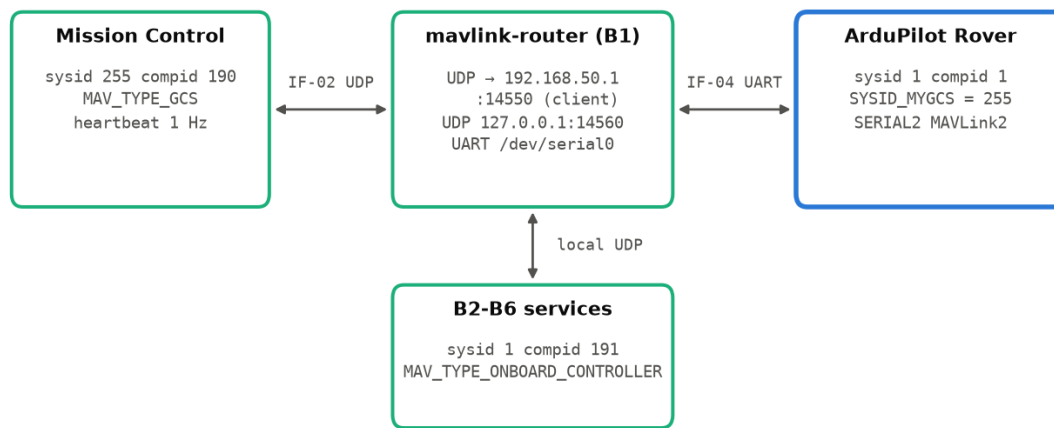


Figure 2. MAVLink identities and routing (IF-02, IF-04).

Identities

Node	System ID	Component ID	MAV_TYPE
ArduPilot Rover (helm)	1	1 (AUTOPILOT1)	SURFACE_BOAT (11)
Mission Control	255	190 (MISSIONPLANNER)	GCS (6)
Mission computer services	1	191 (ONBOARD_COMPUTER)	ONBOARD_CONTROLLER (18)
QGroundControl backup (MC-014)	255	190	GCS (6)

The helm is configured with `SYSID_MYGCS = 255`, so only heartbeats from system 255 count as ground-station presence for the GCS failsafe. The mission computer's heartbeats (system 1, component 191) do not mask a lost bank link (V-06). Only one GCS is connected at a time.

Telemetry the helm sends (requested with `MAV_CMD_SET_MESSAGE_INTERVAL, 511`)

Message (ID)	Rate	Used for
HEARTBEAT (0)	1 Hz	Mode (custom_mode), armed flag, link presence
SYS_STATUS (1)	1 Hz	Sensor health, battery summary
GLOBAL_POSITION_INT (33)	4 Hz	Map position, heading, speed (MC-005)
GPS_RAW_INT (24)	1 Hz	Fix type, satellites, HDOP (PRE-001)
BATTERY_STATUS (147)	1 Hz	Voltage, current, % remaining; instance 1 = motor rail (V-13)
VFR_HUD (74)	2 Hz	Ground speed, throttle
MISSION_CURRENT (42)	1 Hz	Progress; mission time remaining
MISSION_ITEM_REACHED (46)	Event	Photo points (also consumed by MCP)
NAV_CONTROLLER_OUTPUT (62)	1 Hz	Cross-track error (NAV-005)
EKF_STATUS_REPORT (193)	1 Hz	Estimator health (FS-004)
FENCE_STATUS (162)	1 Hz	Breach state and count
STATUSTEXT (253)	Event	Failsafe and pre-arm reasons, turned into plain language (MOD-004)

Commands Mission Control sends

Action	Message / command	Parameters	Allowed when
Arm	COMMAND_LONG: COMPONENT_ARM_DISARM (400)	p1 = 1	State APPROVED, adult key on, pre-arm OK
Disarm	COMMAND_LONG 400	p1 = 0	HOLD at home or after STOP
Force disarm	COMMAND_LONG 400	p1 = 0, p2 = 21196	Only if normal disarm refused after STOP
Set mode	COMMAND_LONG: DO_SET_MODE (176)	p1 = 1 (custom), p2 = mode number	See mode table

Action	Message / command	Parameters	Allowed when
Enable fence	COMMAND_LONG: DO_FENCE_ENABLE (207)	p1 = 1	After fence upload, before arming
Stream rates	COMMAND_LONG: SET_MESSAGE_INTERVAL (511)	p1 = msg id, p2 = interval μ s	On connect
Manual drive	MANUAL_CONTROL (69)	x = throttle, r = turn (-1000..1000), 10 Hz	MANUAL (STEERING) only; adult
Parameter read	PARAM_REQUEST_LIST (21) / PARAM_VALUE (22)	-	On connect (SAF-007 baseline check)
Parameter write	PARAM_SET (23)	-	Disarmed only ; refused by C7 when armed (SAF-003)

Rover mode numbers used (custom_mode)

SRS mode	ArduPilot mode	Number
MANUAL	STEERING	3
HOLD (motors off)	HOLD	4
HOLD (station-keeping)	LOITER	5
(weed-shedding only, IF-04)	GUIDED	15
AUTO	AUTO	10
RTL	RTL	11

Mode numbers follow ArduPilot Rover's published list [\[TBC-01\]](#).

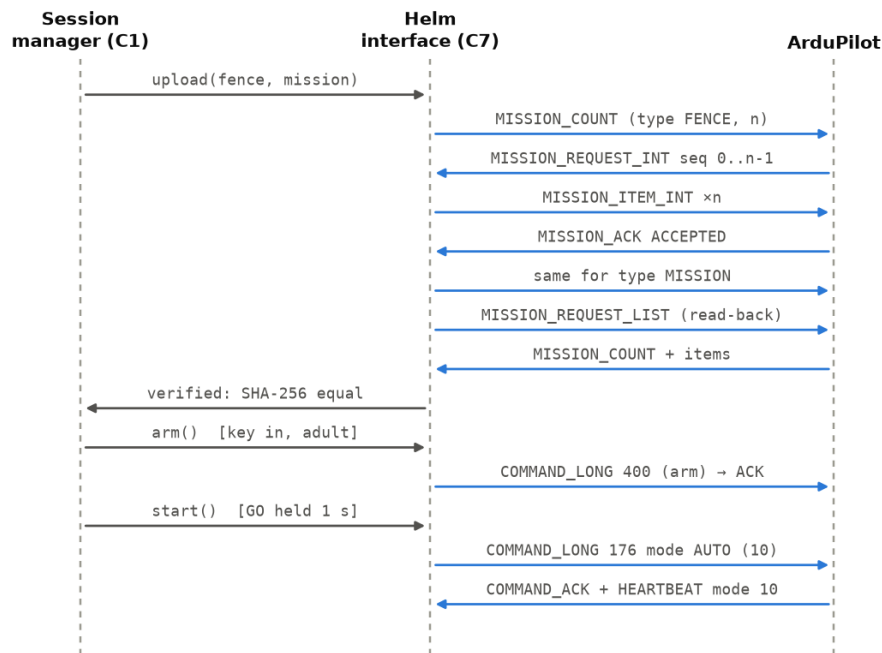


Figure 3. Fence and mission upload, read-back, arming and start (VAL-010, MOD-003).

Mission and fence transfer

- Standard MAVLink mission protocol, integer items: MISSION_COUNT (44), MISSION_REQUEST_INT (51), MISSION_ITEM_INT (73), MISSION_ACK (47). mission_type 1 = fence, 0 = mission. The fence is uploaded first.
- Fence items: FENCE_POLYGON_VERTEX_INCLUSION (5001, p1 = vertex count), FENCE_POLYGON_VERTEX_EXCLUSION (5002), FENCE_CIRCLE_EXCLUSION (5004, p1 = radius m), FENCE_RETURN_POINT (5000) = home.

- Mission items: seq 0 is home (ArduPilot convention; overwritten by the helm on arming). Then items as mapped in IF-13, in MAV_FRAME_GLOBAL_RELATIVE_ALT_INT (6), altitude 0. The last item is NAV_RETURN_TO_LAUNCH (20).
- Read-back: MISSION_REQUEST_LIST (43) per type. The items returned are canonicalised (IF-13) and hashed. GO is enabled only if the hash equals the approved mission's (VAL-010). Seq 0 (home) is excluded from the hash.

Timing and error handling

Case	Rule
COMMAND_ACK	Expected ≤ 1 s. Up to 3 attempts at 1 s intervals, then HelmError('no response') (IF-14).
MAV_RESULT $\neq$ ACCEPTED (0)	Raise CommandRejected with the result code and the latest STATUSTEXT, spoken in plain words (MOD-004).
Mission transfer	Per-item timeout 1.5 s; ≤ 3 retries per item; whole transfer ≤ 20 s or abort and report.
STOP latency budget	Button $\rightarrow$ C1 ≤ 50 ms; C7 $\rightarrow$ radio ≤ 50 ms; link ≤ 300 ms; helm mode change ≤ 100 ms. Total ≤ 0.5 s, against FS-009's 1 s.
Heartbeat loss	No HEARTBEAT from system 1 for 3 s $\rightarrow$ 'link lost' in the UI; keep resending commands only if the user presses again.

Verification

Method	Stage	Evidence / pass criterion
Simulation	SIM	Automated SITL suite: every command in the table accepted or rejected as specified; read-back hash matches; mode numbers confirmed
Test	POOL	STOP $\rightarrow$ motors stopped ≤ 1 s, measured on video with an on-screen timestamp (FS-009)

IF-04 Companion link

Interface	IF-04 Companion link	Owner	MCP (Mission computer)
Between	Mission computer $\leftrightarrow$ Helm	Type	Data
Medium	MAVLink 2 over UART (FC TELEM port)	Requirements	FS-003/006, CAM-003

Physical

Parameter	Value
Signal levels	3.3 V TTL UART, both ends (no level shifting)
Wiring	FC TX $\rightarrow$ Pi GPIO15 (RXD, pin 10); FC RX $\leftarrow$ Pi GPIO14 (TXD, pin 8); GND $\leftrightarrow$ pin 6. No power conductor (each side is powered separately, IF-07/IF-08).
Port	FC SERIAL2 (TELEM2 pads) [TBC-02]
Settings	115200 baud, 8N1, no flow control. SERIAL2_PROTOCOL = 2 (MAVLink 2).
Pi side	/dev/serial0 on the PL011 UART (dtoverlay=disable-bt), serial console disabled
Cable	≤ 200 mm, inside the electronics box

115200 baud carries ≈ 11 kB/s against ≈ 4 kB/s of expected traffic, so the link runs at under 40% load.

mavlink-router configuration (B1)

```

[General]
TcpServerPort = 0          # no TCP server
ReportStats = false

[UartEndpoint helm]
Device = /dev/serial0
Baud = 115200

[UdpEndpoint missioncontrol]
Mode = Normal              # client: sends to the bank
Address = 192.168.50.1
Port = 14550

[UdpEndpoint local]
Mode = Server              # B2-B6 connect here
Address = 127.0.0.1
Port = 14560

```

What boat-side services (component 191) may send

Service	Messages / commands	Constraint
B4 link watchdog	DO_SET_MODE → RTL (11)	Only if mode is AUTO and no system-255 HEARTBEAT for 60 s (FS-003)
B5 weed-shedding	DO_SET_MODE → GUIDED (15); SET_POSITION_TARGET_LOCAL_NED (84) velocity-only in MAV_FRAME_BODY_NED (8); DO_SET_MODE → HOLD (4) or back to the previous mode	Only after a helm stuck event. ≤ 0.5 m/s astern, ≤ 2 s per burst, ≤ 3 bursts, targets at 10 Hz. Relies on V-11.
B6 health	DO_SET_MODE → RTL (11); STATUSTEXT	Moisture detected
B2 camera	None (listens only)	Reads GLOBAL_POSITION_INT, MISSION_ITEM_REACHED, SYSTEM_TIME
All	Never: arm/disarm, PARAM_SET, fence or mission upload, AUTO, MANUAL, STEERING	Enforced by a message filter in the services' shared MAVLink client library

A boat-side request that the helm rejects is logged and not retried more than twice.

Verification

Method	Stage	Evidence / pass criterion
Inspection	BENCH	Wiring and serial settings against this section; loopback test at 115200 with no errors over 10 min
Simulation	SIM	B4, B5 and B6 behaviours in SITL, including killing B5 mid-burst (V-11)
Test	SIM	The filter library refuses every forbidden command (unit tests)

IF-03 Photo & health service

Interface	IF-03 Photo & health service	Owner	MCP (Mission computer)
Between	Mission Control ↔ Mission computer	Type	Data
Medium	HTTP/JSON over IF-01	Requirements	CAM-007, FS-010

A small HTTP/JSON API served by the mission computer on TCP 8080. Mission Control is the only client. Every request carries Authorization: Bearer <token>, a 32-byte random token provisioned on both computers at set-up. Timestamps are ISO 8601 UTC. Coordinates are WGS84 decimal degrees.

Endpoint	Purpose	Response
GET /v1/health	Health snapshot	200 Health
PUT /v1/session	Set the capture policy for the next mission	200 Session
GET /v1/photos?mission=<id>&since=<utc>	List photos	200 [Photo]
GET /v1/photos/<id>	Full image	200 image/jpeg

Endpoint	Purpose	Response
GET /v1/photos/<id>/thumb	320 px thumbnail	200 image/jpeg
POST /v1/photos/ack	Confirm photos safely copied (allows later deletion)	204
POST /v1/capture	Take photos now: {count 1-10, interval_s 0.5-5}	202
POST /v1/shutdown	Clean shutdown before power-off (R-04)	202

```

Health = {"status": "ok"|"degraded"|"fault", "uptime_s": int,
          "cpu_temp_c": float, "throttled": bool, "storage_free_mb": int,
          "photos_unsynced": int, "moisture": bool, "camera_ok": bool,
          "mavlink_ok": bool, "rssi_dbm": int, "sw_version": str}
Session = {"mission_id": uuid, "interval_s": 0 | 2..30,
           "photo_points": [{"seq": int, "burst_n": 1..10}]}
Photo = {"id": str, "mission_id": uuid, "taken_utc": str,
         "lat": float, "lon": float, "heading_deg": float,
         "trigger": "interval"|"photo_point"|"command",
         "point_seq": int|null, "bytes": int, "sha256": str,
         "width": int, "height": int}
Error = {"error": "bad_request"|"not_found"|"busy"|"storage_full",
         "message": str} # with 4xx/5xx status

```

- Images: JPEG, 2592 × 1944 (5 MP), quality 85, ≈ 1.5-2.5 MB. Geotags come from the latest GLOBAL_POSITION_INT, which must be under 0.5 s old, otherwise lat/lon are null.
- Capture triggers: every interval_s during AUTO. At a photo point, burst_n photos starting ≤ 0.5 s after MISSION_ITEM_REACHED for that seq. On POST /v1/capture.
- Sync: thumbnails first, then full images. 200 photos in ≤ 5 min near the bank (CAM-007) needs ≈ 11 Mbit/s. Mission Control verifies each sha256 before calling ack.
- Photos are never deleted before ack. With storage full the service stops capturing and returns 507 storage_full; navigation is unaffected (CAM-004).

Verification

Method	Stage	Evidence / pass criterion
Test	BENCH	Contract tests: every endpoint, error and auth case; the SIM camera stub passes the same suite (IF-21)
Test	POOL	200-photo sync ≤ 5 min with the boat ≤ 10 m from the bank; hashes verified

IF-10 Internet uplink

Interface	IF-10 Internet uplink	Owner	MCN (Mission Control)
Between	Mission Control ↔ Phone	Type	Data
Medium	USB tethering (RNDIS/NCM)	Requirements	COM-005

Parameter	Value
Physical	Phone USB cable to a Pi 5 USB port; phone set to 'USB tethering' (Android) or 'Personal Hotspot' with USB (iPhone)
Link	RNDIS/NCM (Android) or ipheth (iPhone), seen as usb0; address by DHCP from the phone (typically 192.168.42.x Android, 172.20.10.x iPhone)
Routing	Default route via usb0 only. No forwarding between usb0 and wlan1 (ip_forward = 0), so nothing on the boat network can reach the internet.
Allowed outbound	HTTPS 443 to api.anthropic.com; NTP (optional). Everything else is blocked by nftables.
Inbound from phone	HTTP 8000 (web UI) only
Loss of uplink	NL planning and cloud photo analysis are unavailable. The UI shows 'no internet: templates only' (MC-011). Boat link unaffected (COM-005).

Verification

Method	Stage	Evidence / pass criterion
Test	BENCH	With both phone types if available: tether up, API reachable, boat network cannot reach the internet, unplugging the phone leaves the boat link intact

IF-11 Claude API

Interface	IF-11 Claude API	Owner	MCN (Mission Control)
Between	Mission Control ↔ Claude API	Type	Data
Medium	HTTPS, Messages API with tool use	Requirements	NLI-003/004, LOG-004

Mission Control calls the Claude API through the official Anthropic Python SDK (`anthropic`). Two request types. Both use structured outputs, so the response is schema-valid JSON rather than free text. The planner and the validator check it again anyway.

Field	Mission intent request	Photo log request (opt-in)
Endpoint	POST /v1/messages	POST /v1/messages
Model	claude-opus-5	claude-opus-5
Thinking / effort	adaptive / medium (tune against NLI-007)	adaptive / medium
max_tokens	16000	16000
Output	output_config.format = JSON schema of Intent v1 (IF-13)	JSON schema of PhotoLog v1
System prompt	Fixed, versioned, cached (cache_control). Role, the intent vocabulary, safety rules, child-friendly summary style.	Fixed, versioned, cached
User content	Site context JSON (names of areas and landmarks, sizes, battery %, limits) + transcript, clearly delimited as data	Up to 20 thumbnails (base64 JPEG, long edge 1024 px) + photo IDs
Refusal fallback	fallbacks = 'default' (beta server-side-fallback-2026-07-01)	Same
Client timeout / retries	20 s / SDK default 2 retries (429, 5xx, connection)	60 s / 2

Response handling

Outcome	Action
stop_reason = end_turn	Parse JSON, validate against Intent v1 (pydantic), pass to the planner
Schema or planner failure	Retry once, with the error appended as data; then offer templates (NLI-004)
stop_reason = refusal (after fallbacks)	Say 'I can't plan that one - shall we pick an adventure?' and offer templates
stop_reason = max_tokens	Retry once, then templates
HTTP error after SDK retries / timeout	Templates; log the request ID

Data minimisation (LOG-004)

- **Never sent:** audio (speech is transcribed on the Pi 5), coordinates (the model sees named areas and landmarks only; the planner owns geometry), the API key (stored only in the Mission Control keyring, NLI-008).
- Photos are sent only when an adult has enabled cloud analysis for the session. Otherwise the captain's log uses on-device heuristics only.
- Each request's ID, token usage and cost estimate are logged (LOG-002).

Verification

Method	Stage	Evidence / pass criterion
Test	SIM	Recorded evaluation set of ≥ 30 instructions (including ones that must be declined): 100% schema-valid; every must-decline declined; plan shown ≤ 20 s on mobile data (NLI-003..007)
Inspection	SIM	Captured request bodies contain no coordinates or audio; photo requests only with cloud analysis enabled

5. Software contracts

IF-13 Mission format

Interface	IF-13 Mission format	Owner	MCN (Mission Control)
Between	Mission Control	Type	Software
Medium	JSON schema v1 (intent $\rightarrow$ mission $\rightarrow$ MAVLink items)	Requirements	MIS-004, VAL-*

Three JSON documents, each carrying a schema field for versioning. A breaking change bumps the major version. Readers reject unknown major versions. The schemas are kept as pydantic models in the repository; this section is their normative summary.

Intent v1: produced by the Claude API, consumed by the planner

```
{
  "schema": "boaty.intent/1",
  "summary_for_child": str (<= 140 chars, spoken aloud),
  "speed": "slow" | "normal",
  "steps": [ 1..8 of:
    { "op": "explore",      "area": , "coverage": "light"|"medium"|"thorough" }
    { "op": "visit",       "landmark": , "photos": 1..10, "hold_s": 5..60 }
    { "op": "photo_stops", "n": 1..6, "near": | null }
    { "op": "lap",         "area": }
    { "op": "return_home" } # must be the last step
  ],
  "declined": null | { "reason_for_child": str } # set instead of steps
}
```

Names must match the site file (IF-15), by name or alias. Anything else is a planner failure.

Mission v1: produced by the planner, checked by the validator, uploaded by the helm interface

```
{
  "schema": "boaty.mission/1", "id": uuid, "created_utc": str,
  "source": "voice"|"text"|"template"|"map", "site": str,
  "site_version": ,
  "home": { "lat": float, "lon": float },
  "cruise_mps": 0.6..1.2,
  "items": [ { "seq": 1.., "kind": "waypoint"|"photo_point"|"speed"|"rtl",
    "lat": float|null, "lon": float|null,
    "hold_s": int|null, "photos": int|null, "speed_mps": float|null } ],
  "estimates": { "distance_m": float, "duration_s": int, "energy_wh": float },
  "capture": { "interval_s": 0|2..30 },
  "checksum": "sha256:"
}
```

kind	MAVLink item (IF-02)	Fields
waypoint	NAV_WAYPOINT (16)	x = lat $\times 1e7$, y = lon $\times 1e7$; p2 acceptance radius = 3 m (NAV-004)
photo_point	NAV_LOITER_TIME (19)	p1 = hold_s; position as above; photos $\rightarrow$ Session.photo_points (IF-03)
speed	DO_CHANGE_SPEED (178)	p2 = speed_mps (≤ 1.5 , NAV-003)
rtl	NAV_RETURN_TO_LAUNCH (20)	Must be last (MIS-002)

Canonical form and checksum: the items list, serialised as JSON with sorted keys and no whitespace. lat/lon are rendered as integers $\times 1e7$ and floats rounded to 2 decimal places. The checksum is its SHA-256. The helm interface computes the same form from the read-back items (VAL-010).

ValidationResult v1: produced by the validator

```
{
  "schema": "boaty.validation/1",
  "mission_id": "uuid",
  "ok": bool,
  "violations": [
    {
      "rule": "VAL-002",
      "message_for_adult": str,
      "message_for_child": str,
      "item_seq": int | null,
      "at": {
        "lat": float,
        "lon": float
      } | null
    }
  ],
  "checked_utc": str,
  "validator_version": str
}
```

Verification

Method	Stage	Evidence / pass criterion
Test	SIM	Schema round-trip tests; validator suite $\geq$ 95% branch coverage with adversarial missions (VAL-006); checksum equal for upload and read-back in SITL

IF-14 Helm API

Interface	IF-14 Helm API	Owner	MCN (Mission Control)
Between	Mission Control	Type	Software
Medium	Python abstract class	Requirements	SWE-002/003

The only way Mission Control code commands a helm (SAF-003, SWE-002/003). Mk1 has one implementation, ArduPilotHelm (pymavlink over IF-02). A future Python helm implements the same protocol.

```
class Helm(Protocol):
    def connect(self, timeout_s: float = 10) -> None
    def status(self) -> HelmStatus # latest snapshot, never blocks
    def subscribe(self, cb: Callable[[HelmEvent], None]) -> None

    def upload_fence(self, fence: Fence) -> None # disarmed only
    def upload_mission(self, m: Mission) -> None # disarmed only
    def read_back(self) -> tuple[Fence, list[MissionItem]]
    def verify(self, m: Mission) -> bool # checksum equality (VAL-010)

    def arm(self) -> None # raises PreArmFailed(reasons)
    def disarm(self, force: bool = False) -> None
    def start_mission(self) -> None # AUTO
    def hold(self) -> None # LOITER: station-keeping
    def stop(self) -> None # HOLD (motors off), then disarm
    def return_home(self) -> None # RTL
    def manual(self) -> None # STEERING; adult only
    def drive(self, throttle: float, turn: float) -> None # -1..1, send  $\geq$  5 Hz

@dataclass(frozen=True)
class HelmStatus:
    t_utc: datetime; link_ok: bool; armed: bool; mode: SrsMode
    lat: float | None; lon: float | None; heading_deg: float | None
    speed_mps: float; battery_pct: float; battery_v: float; rail_v: float
    gps_fix: int; sats: int; hdop: float; ekf_ok: bool; fence_breached: bool
    mission_seq: int | None; mission_total: int | None

HelmEvent = ModeChanged | FailsafeEvent | ArrivedHome | Breach | Text | LinkChange
Exceptions: HelmError > {NoResponse, CommandRejected(code, text), PreArmFailed(reasons),
    NotAllowedWhileArmed, TransferFailed}
```

- Thread-safe. Blocking calls honour the IF-02 timeouts. status() is served from a cache refreshed by the telemetry stream.
- stop() is idempotent and has the highest priority: it pre-empts any call in progress.
- Upload and parameter calls raise NotAllowedWhileArmed when armed.
- SrsMode maps from ArduPilot modes as in the IF-02 table. GUIDED appears only during weed-shedding and is shown as HOLD.

Verification

Method	Stage	Evidence / pass criterion
Test	SIM	Contract test suite run against ArduPilotHelm on SITL; the same suite is the acceptance test for any future helm

IF-15 Site data

Interface	IF-15 Site data	Owner	MCN (Mission Control)
Between	Mission Control ↔ Simulation & test	Type	Software
Medium	GeoJSON in git	Requirements	FEN-001/002

One GeoJSON (RFC 7946) FeatureCollection per site, in `sites/<site>.geojson`, committed to git (FEN-002). Coordinates are [lon, lat] in WGS84.

properties.role	Geometry	Other properties	Rules
fence_inclusion	Polygon	-	Exactly one; ≤ 70 vertices; drawn ≥ 5 m inside the waterline (FEN-003)
exclusion	Polygon or Point	radius_m (Point), reason	≤ 10 in total; inside the inclusion (FEN-001, OPS-005)
home	Point	name	≥ 1 ; inside the inclusion, outside exclusions (PRE-002)
area	Polygon	name, aliases[]	e.g. 'home bay'; inside the inclusion
landmark	Point or Polygon	name, aliases[], keep_out_m	e.g. 'the island' with keep_out_m = 10; photo stops are placed at $\geq$ keep_out_m

```
{
  "type": "FeatureCollection",
  "properties": {
    "site": "milton-todds-pit",
    "version": 3,
    "wifi_channel": 6,
    "max_distance_from_home_m": 100
  },
  "features": [
    {
      "type": "Feature",
      "properties": {
        "role": "home",
        "name": "jetty"
      },
      "geometry": {
        "type": "Point",
        "coordinates": [0.16xx, 52.24xx]
      }
    },
    ...
  ]
}
```

Coordinates are shown as placeholders. The real site file is drawn on the map once TBD-02 (which lake) is resolved.

Verification

Method	Stage	Evidence / pass criterion
Test	SIM	Site-file linter: schema, geometric rules above, and fence round-trip through IF-02 in SITL

IF-21 Simulation link

Interface	IF-21 Simulation link	Owner	SIM (Simulation & test)
Between	Simulation & test ↔ Mission Control	Type	Data
Medium	MAVLink 2 UDP from SITL; camera stub with the IF-03 API	Requirements	SWE-004/005

```
sim_vehicle.py -v Rover -f motorboat-skid \
  --custom-location=,,0, \
  --add-param-file=params/boaty-mk1.parm --out=udp:127.0.0.1:14550
python -m boaty.sim.camera_stub --port 8080 --images tests/images/ducks/
```

- SITL presents the same MAVLink as IF-02 on UDP 14550. The camera stub implements IF-03 exactly and reads positions from SITL, so Mission Control runs unmodified with its host set to 127.0.0.1 (SWE-004).
- The frame name is **TBC-03**. The parameter file is the same controlled file flashed to the real helm, plus `SIM_*` overrides.
- Fault injection for SWE-005 uses SITL simulation parameters (GPS failure, battery drain, added drag) and network impairment (dropping UDP 14550) **TBC-04**.

Verification

Method	Stage	Evidence / pass criterion
Demonstration	SIM	A full mission from voice to captain's log in simulation; every FS scenario runs in CI

6. Electrical interfaces

IF-05 Motor commands

Interface	IF-05 Motor commands	Owner	HLM (Helm)
Between	Helm ↔ Propulsion	Type	Data
Medium	2 × DShot or PWM (bidirectional)	Requirements	NAV-001, FS-008

Parameter	Value
Protocol	DSHOT300, bidirectional ('3D') ESC mode. Fallback: PWM 1000-2000 μ s, 50 Hz, neutral 1500 μ s, deadband $\pm$ 25 μ s [TBC-05]
Channel map	FC output 1 = left (ThrottleLeft), output 2 = right (ThrottleRight)
Sense	Positive = forward thrust on both sides
Wiring	Signal + GND twisted pair, $\leq$ 150 mm; no power from the ESCs to the FC
Loss of signal	ESC stops the motor $\leq$ 1 s after pulses stop (FS-008, V-09)
Update	Helm output loop $\geq$ 50 Hz

Verification

Method	Stage	Evidence / pass criterion
Test	BENCH	Direction and mapping with props off; signal-loss stop time measured by pulling the signal lead (V-09)

IF-06 Motor power

Interface	IF-06 Motor power	Owner	PWR (Power)
Between	Power ↔ Propulsion	Type	Power
Medium	11.1 V nominal via key-switched MOSFET, XT30	Requirements	PWR-003/008, MOD-003

Parameter	Value
Voltage	3S Li-ion: 9.0-12.6 V operating (BMS cut-off $\approx$ 7.5 V)
Current	$\leq$ 8 A continuous per ESC (estimate), $\leq$ 25 A total peak for 100 ms [TBC-06]
Protection	20 A blade fuse $\leq$ 50 mm from the battery positive (PWR-003)
Key switch	Normally-open reed switch on the lid. Magnet present $\rightarrow$ MOSFET on $\rightarrow$ motor rail live. Magnet absent $\rightarrow$ gate pulled down $\rightarrow$ rail off (fails safe). MOSFET $\geq$ 40 A, $R_{DS(on)} \leq$ 10 m Ω .
Rail sense	Divider 10 k Ω / 1 k Ω to the FC's second voltage input ($\leq$ 1.2 V at 12.6 V). Arming blocked below 9.0 V [TBC-07]
Connectors	XT30 per ESC; battery XT60; polarised, with no same-type connector used for another function (PWR-008)
Wire	16 AWG battery $\rightarrow$ bus; 18 AWG bus $\rightarrow$ ESC; silicone insulation

Verification

Method	Stage	Evidence / pass criterion
Test	BENCH	Key out $\rightarrow$ rail $<$ 0.5 V within 100 ms; key in $\rightarrow$ rail = battery – 0.2 V; arming refused with key out; fuse and connector inspection

IF-07 Helm power & sensing

Interface	IF-07 Helm power & sensing	Owner	PWR (Power)
Between	Power ↔ Helm	Type	Power
Medium	Power module: supply + analogue V/I; motor-rail sense	Requirements	PWR-005/006, PRE-003

Parameter	Value
Supply	FC powered from the main bus (after the fuse and main switch, before the key) via its own 3-6S input regulator
Sensing	Analogue voltage and current from the power module to the FC battery-monitor pins; scale factors calibrated on the bench
Accuracy	Voltage $\pm 1\%$, current $\pm 5\%$ after calibration; mAh consumed used for SoC (FS-001)
Also	Second monitor = motor-rail sense (IF-06)

Verification

Method	Stage	Evidence / pass criterion
Test	BENCH	Compare with a multimeter at 3 loads; SoC tracks a measured discharge within 5%

IF-08 Mission computer power

Interface	IF-08 Mission computer power	Owner	PWR (Power)
Between	Power ↔ Mission computer	Type	Power
Medium	Dedicated 5 V 3 A buck	Requirements	PWR-005

Parameter	Value
Source	Main bus before the key switch, so the MCP can sync photos with motors off
Regulator	Buck, 5.1 V $\pm 2\%$, ≥ 3 A continuous, ripple ≤ 50 mV p-p, input 7-14 V
Connection	Pi GPIO header pin 2 (5 V) and pin 6 (GND), keyed 2-pin connector
Transients	Output stays ≥ 4.85 V through a full reverse-to-forward thrust step (PWR-005)
Shutdown	POST /v1/shutdown (IF-03) before the main switch is turned off; the root filesystem is read-only, so an unplanned power cut is survivable (R-04)

Verification

Method	Stage	Evidence / pass criterion
Test	BENCH	Scope the 5 V rail during thrust steps; 50 power-cut cycles without filesystem damage

IF-09 Beacon drive

Interface	IF-09 Beacon drive	Owner	HLM (Helm)
Between	Helm ↔ Recovery & signalling	Type	Data
Medium	Notify-LED output (WS2812-type)	Requirements	REC-004

Parameter	Value
Device	WS2812-type ring, 8 LEDs, at the mast head
Drive	FC output configured as NeoPixel. 330 Ω series resistor. Data cable ≤ 400 mm inside the mast tube.
Power	5 V from the FC's 5 V pad, brightness capped so the draw is ≤ 250 mA
Patterns	ArduPilot notify patterns: armed, failsafe, low battery. Documented in the helm spec after bench observation (REC-004).
Level	3.3 V data into 5 V LEDs [TBC-08]

Verification

Method	Stage	Evidence / pass criterion
Test	BENCH / LAKE	Each state shows its pattern on the bench; visible at 100 m in overcast daylight (REC-004)

IF-22 GNSS & compass

Interface	IF-22 GNSS & compass	Owner	HLM (Helm)
Between	Helm	Type	Data
Medium	UART (u-blox) + I2C compass	Requirements	PRE-001, NAV-008

Parameter	Value
Module	u-blox M10 GNSS with magnetometer, on the mast head ≥ 150 mm from power wiring (MEC-013)
GNSS	UART at 3.3 V to an FC serial port set to GPS. ArduPilot auto-configures the receiver (baud, 5-10 Hz).
Compass	I2C to the FC; external compass with the orientation set in parameters [TBC-09]
Power	5 V from the FC, ≈ 50 mA
Cable	≤ 450 mm through the mast tube; the module connector is strain-relieved at the mast head

Verification

Method	Stage	Evidence / pass criterion
Test	BENCH	3D fix outdoors, ≥ 8 satellites, HDOP ≤ 1.5 (PRE-001); compass heading within 10° of a reference with the motors running

7. Human-machine interface

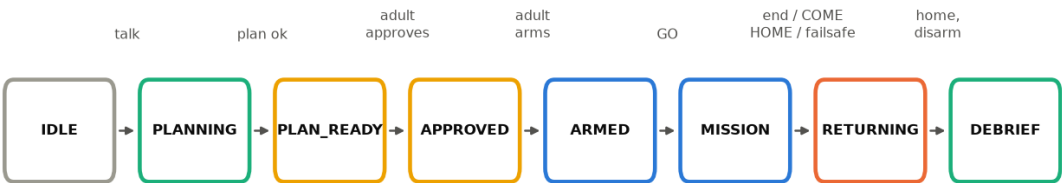
IF-12 Crew & operator panel

Interface	IF-12 Crew & operator panel	Owner	MCN (Mission Control)
Between	Mission Control ↔ Operator & crew	Type	HMI
Medium	4 buttons, key switch, LEDs, mic, speaker, web UI	Requirements	MC-002..008, CHD-005

Panel hardware (Pi 5 GPIO, BCM numbering)

Signal	GPIO (pin)	Electrical	Behaviour
GO button (green, ►)	17 (11)	Input, pull-up, active-low	Press-and-hold 1.0 ± 0.1 s (MC-004)
COME HOME (yellow, house)	27 (13)	Input, pull-up	Acts on press
STOP (red, ■)	22 (15)	Input, pull-up	Acts on press; edge detected ≤ 50 ms
TALK (blue, mic) (TBC-10 closed)	24 (18)	Input, pull-up	Hold to talk, release to send (NLI-002)
Adult key switch	23 (16)	Input, pull-up	On = adult functions unlocked (MC-008)
Button LEDs (4)	5, 6, 13, 19 (29, 31, 33, 35)	Output via N-MOSFET to 5 V LEDs	Lit = available now

Debounce 30 ms in software. Buttons are ≥ 30 mm arcade buttons with 5 V LEDs (MC-002).



STOP from any armed state → motors stop, HOLD, then disarm → IDLE (or DEBRIEF if photos exist). Any edit to an approved plan → PLAN_READY.

Figure 4. Session states in the session manager (C1), which decide what each button does.

Button behaviour by session state

State	GO	COME HOME	STOP	TALK
IDLE / DEBRIEF	'Not yet!'	-	-	Plan
PLAN_READY / APPROVED	'Ask a grown-up to arm'	-	-	Re-plan
ARMED (HOLD)	Start mission	RTL	Stop + disarm	-
MISSION	-	RTL	Stop	-
RETURNING	-	-	Stop	-
MANUAL (adult)	-	RTL	Stop	-

'-' = ignored, with a short 'not now' sound. Button LEDs show which buttons do something right now.

Audio

Item	Value
Microphone	USB, 16 kHz mono; captured only while TALK is held; audio is never stored or sent
Speaker	USB audio, ≈ 3 W; volume set in the UI
Speech in / out	On-device STT and TTS (ADD DD-07)

Event	Spoken phrase (examples)
Plan ready	The plan's summary_for_child, then 'Ask a grown-up to check it'
Mission start	'Off we go!'
Photo point	'Taking pictures!'
COME HOME / RTL	'Coming home!'
STOP	'Stopping!'
Failsafe	Child: 'I need to come home now.' Adult screen: the reason (FS-012).
Stuck	'I'm stuck in some weed, trying to wiggle free.'
Home	'I'm back! Let's look at the pictures.'

Web UI

- Served at <http://192.168.50.1:8000> (tablet on the AP) and on the usb0 address (phone). The address is spoken at start-up.
- Pages: Map (live), Plan (preview and approval), Checklist, Fence editor, Photos, Settings. Adult pages need the key switch on.
- State pushed to the browser over a WebSocket at ≥ 1 Hz (MC-005).

Verification

Method	Stage	Evidence / pass criterion
Test	BENCH	Automated test of the state $\times$ button table with SITL; GO hold timing; STOP detection ≤ 50 ms
Demonstration	POOL	The crew uses GO, COME HOME and STOP without reading (CHD-005)

8. Mechanical interfaces

Dimensions marked TBC are fixed by CAD and test prints and then entered here. Printed parts are PETG unless stated. Load cases use a design mass of 2.1 kg (boat plus 300 g payload) and a safety factor of 3.

IF-16 Hull segment joint

Interface	IF-16 Hull segment joint	Owner	HUL (Hull & structure)
Between	Hull & structure	Type	Mechanical
Medium	Flange + printed thumb-screws	Requirements	MEC-007, CHD-004

Feature	Definition
Section envelope	90 mm wide × 100 mm deep at the joint [TBC-11]
Flange	6 mm thick; mating faces flat to 0.3 mm
Alignment	2 × Ø8 mm printed dowels, 10 mm engagement
Fastening	2 × printed thumb-screws, M12 coarse printed thread, Ø35 mm knurled knob, captive (cannot be fully removed) so there are no small loose parts (CHD-001)
Load case	Boat supported at bow and stern only; the joint carries 3 × design weight without separation or cracking
Sealing	Not required (foam-filled, REC-001). Optional 1 mm EPDM gasket.

Verification

Method	Stage	Evidence / pass criterion
Test	BENCH	Load test on two joined segments; the crew assembles a joint by hand (CHD-004)

IF-17 Pod mount

Interface	IF-17 Pod mount	Owner	PRP (Propulsion)
Between	Propulsion ↔ Hull & structure	Type	Mechanical
Medium	Dovetail + retaining pin; keyed 3-pin waterproof connector	Requirements	MEC-008

Feature	Definition
Mount	60° dovetail, 20 mm wide × 40 mm long, on the hull transom; 0.3 mm clearance
Retention	Spring-loaded printed latch, released by an adult with a firm two-finger press. No separate pin (no loose parts).
Loads	Thrust ≤ 10 N per pod × 3; lateral impact 20 N at the pod tip
Connector	Keyed 3-pole IP68 circular connector (motor phases), ≥ 10 A [TBC-12]
Guard	Prop fully shrouded; an 8 mm probe cannot touch a blade (MEC-010)
Swap time	< 2 min, no tools (MEC-008)

Verification

Method	Stage	Evidence / pass criterion
Test	BENCH	Pull test, probe test, timed swap

IF-18 Box & glands

Interface	IF-18 Box & glands	Owner	HUL (Hull & structure)
Between	Power ↔ Hull & structure	Type	Mechanical
Medium	Saddle on rail; PG7 glands	Requirements	MEC-012

Feature	Definition
Box	Clip-lock food box, internal ≥ 180 × 110 × 70 mm [TBC-13]
Saddle	Printed cradle on the crossbeam rail with 2 over-centre printed latches; adult removal without tools

Feature	Definition
Glands	4 × PG7 (cable Ø3-6.5 mm) on the aft face, 25 mm pitch; hand-tight + ¼ turn
Key dock	Printed cup on the lid over the reed switch; the magnet key on a lanyard; lid label 'KEY'
Ingress	No water after 30 min at 0.3 m (MEC-012)

Verification

Method	Stage	Evidence / pass criterion
Test	BENCH	Submersion test with tissue indicator inside

IF-19 Mast socket

Interface	IF-19 Mast socket	Owner	HUL (Hull & structure)
Between	Helm ↔ Hull & structure	Type	Mechanical
Medium	Socket on rail; GNSS ≥ 150 mm from power wiring	Requirements	MEC-013, REC-005

Feature	Definition
Tube	Ø16 mm aluminium or carbon; GNSS top ≈ 450 mm above the waterline, total height ≤ 500 mm (MEC-001)
Socket	Printed, 50 mm deep, on the rail; retained by a captive thumb-screw
Carries	GNSS/compass (top), flag (top ≥ 300 mm above the waterline), recovery hoop (Ø ≥ 60 mm internal), LED ring
Loads	50 N tow on the hoop (REC-005) → ≈ 15 N·m at the socket; design for 30 N·m
Separation	GNSS ≥ 150 mm from motor and power wiring (MEC-013)

Verification

Method	Stage	Evidence / pass criterion
Test	POOL	50 N hoop pull; tow the boat by the hoop without capsize

IF-20 DUPLO deck

Interface	IF-20 DUPLO deck	Owner	HUL (Hull & structure)
Between	Hull & structure ↔ Operator & crew	Type	Mechanical
Medium	DUPLO-compatible stud grid	Requirements	MEC-009

Feature	Definition
Grid	DUPLO-compatible studs, 16.0 mm pitch, 6 × 8 studs (96 × 128 mm)
Stud	Nominal Ø9.4 mm × 4.5 mm high [TBC-14]
Grip	A 2×2 brick stays on at 45° tilt and with 0.5 g shake. A 4-year-old can remove it (MEC-009).
Mount	Clips onto the box-lid frame; removable for box access

Verification

Method	Stage	Evidence / pass criterion
Test	BENCH	Tilt and shake test; crew removal test

9. Items to be confirmed

Known gaps. Each closes by the method shown and is then entered in the relevant section in the next issue.

ID	Item	Interface
TBC-01	Confirm Rover mode numbers against the firmware version flashed (SITL check).	IF-02
TBC-02	Confirm which UART is free on the chosen F405 board.	IF-04
TBC-03	Confirm a skid-steer boat frame name in the SITL version used (V-12).	IF-21
TBC-04	Map each FS scenario to specific SIM_* parameters for the SITL version used.	IF-21
TBC-05	Confirm the chosen ESC supports DShot and 3D mode.	IF-05
TBC-06	Measure motor current on the bench and re-rate.	IF-06
TBC-07	Confirm arming can be gated on the second battery monitor (V-13).	IF-06
TBC-08	Check reliable data at 3.3 V; add a level shifter if not.	IF-09
TBC-09	Confirm the compass chip, I2C address and orientation for the chosen module.	IF-22
TBC-10	Closed. Owner approved the TALK button; MC-002 now names four buttons (SRS Issue D) (was: Owner to confirm a fourth 'TALK' button: MC-002 names three, but push-to-talk (NLI-002) needs a trigger a 4-year-old can find.)	IF-12
TBC-11	Final hull lines from CAD.	IF-16
TBC-12	Select the connector part number.	IF-17
TBC-13	Choose the box; freeze the saddle.	IF-18
TBC-14	Tune stud diameter and height by test prints against genuine bricks.	IF-20

10. Interface verification summary

Interface	Method	Stage	Evidence / pass criterion
IF-01	Test	BENCH	Association, addressing and firewall rules checked with a script; ports reachable only as listed
IF-01	Test	LAKE	≥ 100 m over water with telemetry ≥ 1 Hz and command latency ≤ 1 s at the 95th percentile (COM-002/003, V-08)
IF-02	Simulation	SIM	Automated SITL suite: every command in the table accepted or rejected as specified; read-back hash matches; mode numbers confirmed
IF-02	Test	POOL	STOP $\rightarrow$ motors stopped ≤ 1 s, measured on video with an on-screen timestamp (FS-009)
IF-04	Inspection	BENCH	Wiring and serial settings against this section; loopback test at 115200 with no errors over 10 min
IF-04	Simulation	SIM	B4, B5 and B6 behaviours in SITL, including killing B5 mid-burst (V-11)
IF-04	Test	SIM	The filter library refuses every forbidden command (unit tests)
IF-03	Test	BENCH	Contract tests: every endpoint, error and auth case; the SIM camera stub passes the same suite (IF-21)
IF-03	Test	POOL	200-photo sync ≤ 5 min with the boat ≤ 10 m from the bank; hashes verified
IF-10	Test	BENCH	With both phone types if available: tether up, API reachable, boat network cannot reach the internet, unplugging the phone leaves the boat link intact
IF-11	Test	SIM	Recorded evaluation set of ≥ 30 instructions (including ones that must be declined); 100% schema-valid; every must-decline declined; plan shown ≤ 20 s on mobile data (NLI-003..007)
IF-11	Inspection	SIM	Captured request bodies contain no coordinates or audio; photo requests only with cloud analysis enabled
IF-13	Test	SIM	Schema round-trip tests; validator suite $\geq 95\%$ branch coverage with adversarial missions (VAL-006); checksum equal for upload and read-back in SITL
IF-14	Test	SIM	Contract test suite run against ArduPilotHelm on SITL; the same suite is the acceptance test for any future helm
IF-15	Test	SIM	Site-file linter: schema, geometric rules above, and fence round-trip through IF-02 in SITL

Interface	Method	Stage	Evidence / pass criterion
IF-21	Demonstration	SIM	A full mission from voice to captain's log in simulation; every FS scenario runs in CI
IF-05	Test	BENCH	Direction and mapping with props off; signal-loss stop time measured by pulling the signal lead (V-09)
IF-06	Test	BENCH	Key out → rail < 0.5 V within 100 ms; key in → rail = battery – 0.2 V; arming refused with key out; fuse and connector inspection
IF-07	Test	BENCH	Compare with a multimeter at 3 loads; SoC tracks a measured discharge within 5%
IF-08	Test	BENCH	Scope the 5 V rail during thrust steps; 50 power-cut cycles without filesystem damage
IF-09	Test	BENCH / LAKE	Each state shows its pattern on the bench; visible at 100 m in overcast daylight (REC-004)
IF-22	Test	BENCH	3D fix outdoors, ≥ 8 satellites, HDOP ≤ 1.5 (PRE-001); compass heading within 10° of a reference with the motors running
IF-12	Test	BENCH	Automated test of the state × button table with SITL; GO hold timing; STOP detection ≤ 50 ms
IF-12	Demonstration	POOL	The crew uses GO, COME HOME and STOP without reading (CHD-005)
IF-16	Test	BENCH	Load test on two joined segments; the crew assembles a joint by hand (CHD-004)
IF-17	Test	BENCH	Pull test, probe test, timed swap
IF-18	Test	BENCH	Submersion test with tissue indicator inside
IF-19	Test	POOL	50 N hoop pull; tow the boat by the hoop without capsize
IF-20	Test	BENCH	Tilt and shake test; crew removal test